

MELODY YIYUAN ZHANG

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EDUCATION

Ph.D. in Chemical Engineering and Scientific Computing

University of Michigan, Ann Arbor, MI

Expected May 2027

GPA: 3.81/4.00

B.S. in Chemical Engineering

Purdue University, West Lafayette, IN

May 2022

GPA: 3.82/4.00

SKILLS

Programming & HPC: Python, C++, Bash, Git, MPI, OpenMP, CUDA

Machine Learning: PyTorch, scikit-learn; variational autoencoders, graph neural networks, SBERT

Simulation & Visualization: LAMMPS, HOOMD-blue, OVITO, VMD, Blender

EXPERIENCE

Advanced Modeling and AI Solutions Intern, Micron Technology, Boise, ID *May – Aug. 2026*

- Designed a material search agent that unifies naming conventions across multiple databases and intelligently interprets non-canonical user queries.
- Integrated automated data retrieval with process and structural modeling software, accelerating simulation setup and reducing manual engineering effort.

Graduate Student Research Assistant, Lindsey & Glotzer Groups

Oct. 2022 – Present

Machine Learning Framework for Inverse Design of Polymer-Grafted Nanoparticles

- Developed a machine-learning inverse design framework, enabling direct optimization of nanoparticle physicochemical attributes that drive self-assembly into target structures.
- Implemented a custom alchemical Monte Carlo algorithm in HOOMD-blue to efficiently perform inverse design searches.

Orientation-Aware Machine-Learning Model for Anisotropic Nanoparticles

- Built an orientation-aware machine-learning interaction model capturing anisotropic interactions in colloidal nanoparticles for self-assembly studies.
- Devised an adaptive parallel meshing algorithm for training data generation, reducing required data volume by $\sim 100\times$ by focusing on key features of the target potential energy surface.

Validation Engineering Intern, AstraZeneca, Mt. Vernon, IN

May – July 2021

- Performed product quality analysis on active pharmaceutical ingredients using BIOVIA Discoverant.
- Designed an interactive visualization platform for equipment and process risk assessments.

PUBLICATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Machine-Learning Interaction Model for Anisotropic Interactions. Manuscript in preparation.

Zhang, M. Y.*, Lee, S. K. A.*, Glotzer, S. C., and Lindsey, R. K. A Generalized Machine-Learning Framework for Developing Alchemical Many-Body Interaction Models for Polymer-Grafted Nanoparticles. *J. Chem. Theory Comput.* 2025, 21(19), 9853–9867 (*equal contribution). DOI: 10.1021/acs.jctc.5c00901

Finkenauer, B., **Zhang, Y.**, Ma, K., Turnley, J. W., Schulz, J., et al. Amine-Thiol/Selenol Chemistry for Efficient and Stable Perovskite Solar Cells. *J. Phys. Chem. C* 2023, 127(2), 930–938. DOI: 10.1021/acs.jpcc.2c06967

AWARDS

Empowering Research with AI Award, AI Institutes at Michigan (AIIM) *2026*

Best Overall Poster Award, U-M Annual Data Science & AI Summit *2025*

Michigan Institute for Computational Discovery & Engineering (MICDE) Fellowship *2025*

2nd Place, U Michigan Ross Business+Tech Datathon *2025*

1st Place, College of Engineering Undergraduate Research Conference *2021*

Stephen Craig Outstanding Junior Award, Chemical Engineering *2021*

AstraZeneca Rising Star Award for initiative, innovation, and productivity *2021*

RELEVANT COURSEWORK

Design & Analysis of Algorithms

Machine Learning

Parallel Computing

Applied Math for Chemical Engineering

Numerical Linear Algebra

Big Data Management Tools

Statistical & Irreversible Thermodynamics

Simulation of Condensed Matter